Lesson Planning

Complete Lesson Plan Book

Class: Class 12

Subject: Chemistry

Medium: English

Academic Year 2026-27

This book is prepared as a teaching guide for teachers.

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Unit 1: Chemical Equilibrium: Le Chatelier's Principle

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Unit 1: Chemical Equilibrium: Le Chatelier's Principle

Learning Objectives

• Students will state Le Chatelier's principle in their own words.

• Students will predict the direction of shift when concentration, pressure or temperature changes.

• Students will explain the effect of a catalyst on an equilibrium system.

Key Concepts

• A system at equilibrium shifts to oppose any change imposed on it — this is Le Chatelier's principle.

• Adding a reactant, or removing a product, shifts equilibrium forward; adding a product shifts it backward.

• Increasing pressure shifts equilibrium towards the side with fewer gas molecules; pressure has no effect when both sides have equal gas moles.

• For an exothermic reaction, raising the temperature shifts equilibrium backward; for an endothermic reaction, it shifts forward.

• A catalyst speeds up both forward and reverse reactions equally, so it helps equilibrium arrive faster but does not change its position.

Teaching Methodology (Step by Step)

1. Recall dynamic equilibrium: forward and reverse rates are equal, so concentrations stay constant.

2. State Le Chatelier's principle and apply it to a change in concentration with the Haber process as an example.

3. Work through pressure and temperature changes on the board, one case at a time.

4. Students predict shifts for five new changes in pairs and justify each answer.

5. Students summarise the three kinds of change in a small table in their notebooks.

Classroom Activities

• Shift predictor: groups predict and defend the direction of shift for given changes.

• Haber case study: students suggest conditions that favour more ammonia.

• Myth buster: students explain why a catalyst cannot increase the yield.

Assessment Questions

1. State Le Chatelier's principle.

2. In $N_2 + 3H_2 \rightleftharpoons 2NH_3$, what happens if more hydrogen is added?

3. Why does raising the temperature lower the yield of an exothermic reaction?

Homework

• Predict the shift for the equilibrium $2SO_2 + O_2 \rightleftharpoons 2SO_3$ when pressure is increased and when SO_3 is removed.

• Explain, with reasons, the best temperature and pressure conditions for the Haber process.

Time Allocation

40 minutes